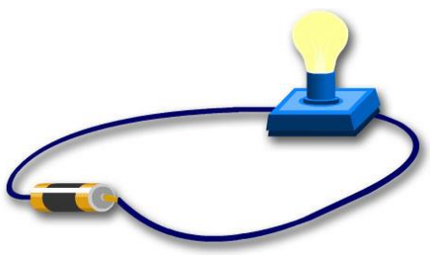
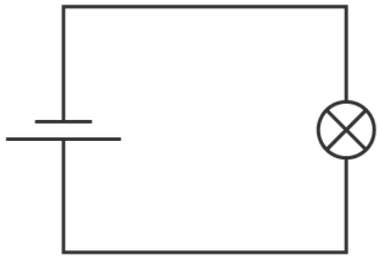
Knowledge Organiser
Year 7: Electricity – Making Circuits

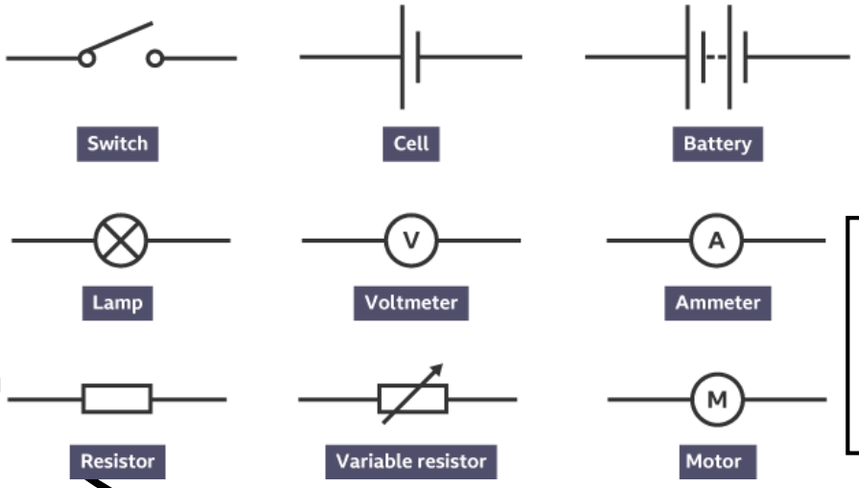
Why are some objects charged?
- A circuit is a path along which electric current flows
- Current is a **flow of electrical energy**, just like when a river flows, energy is transferred through the circuit
- A simple circuit consists of a **power source** (such as a battery), conductive wires, and a component (such as a bulb or motor)

How to draw a circuit
- Use a **pencil and ruler** and draw the circuit using straight lines use symbols to represent the components e.g. a bulb is a circle with a cross through it
- Connect the **components** with straight lines to show how they are connected in the circuit
- Make sure there are **no gaps** in the circuit, we call this a **closed circuit**





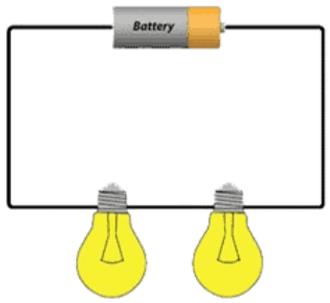
A diagram showing how a simple circuit is converted into a circuit diagram. It uses straight lines and circuit symbols.



Series and Parallel Circuits
- In a series circuit, the components are connected in a **single path**
- The same current flows through each component in a series circuit
- In a parallel circuit, the components are connected in **multiple branches**
- Each component is connected to the same two points in the circuit, creating **parallel paths** for the current to flow

These two bulbs are **in series** with each other.

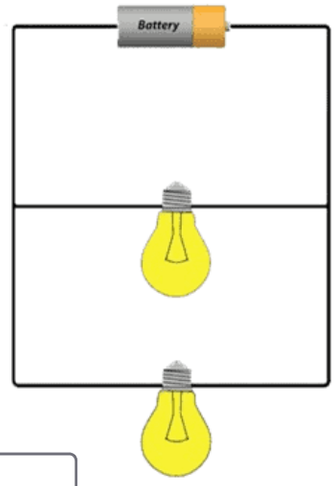
Current must flow through each bulb in sequence.

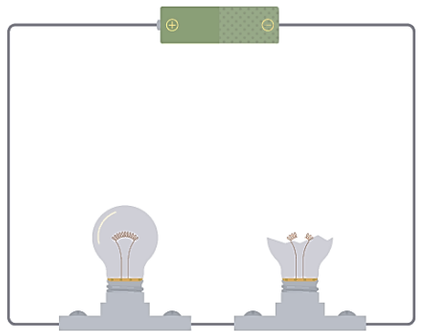


These two bulbs are **in parallel** with each other.

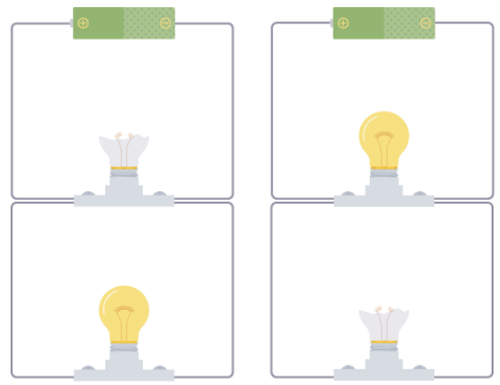
Current can only flow through one “loop” of the circuit at a time.

The current is split between the bulbs.



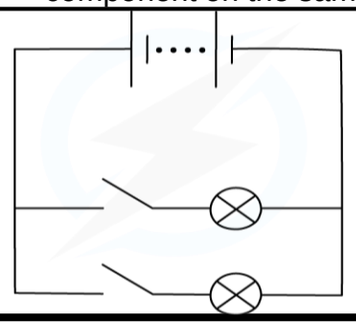


What happens if a series circuit breaks
- If a component in a series circuit breaks, like this bulb then **current cannot flow**
- The circuit is “broken” because **the flow of charged particles** cannot get through the broken bulb and back to the battery



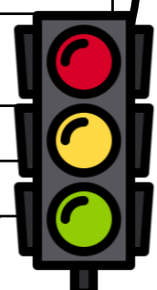
What happens if a parallel circuit breaks
- If a component in a parallel series breaks, then the current will still flow through the other loops
- In the diagram above, the bulb that is not broken is still able to turn on
- The charged particles can still flow from the battery, through the functional bulb and back to the battery

Where do we use parallel circuits?
- Our homes are wired up using parallel circuits
- This way, if one component breaks the others will still keep working
- It also enables us to have separate switches for each component on the same circuit



In the parallel circuit shown in this diagram, the two bulbs can be “switched on” separately by closing one of the switches

Learning Outcomes;	**RAG**
Draw complete circuit diagrams using a rule, pencil and straight lines	
Draw a range of common circuit symbols e.g. wire, voltmeter, bulb	
Describe the difference between a series and parallel circuit	
Explain why current is the same at all points in a series circuit	
State which particle moves in an electrical current	
Explain why parallel circuits are used in the home	



Knowledge Organiser
Year 7: Electricity – Making Circuits

Why are some objects charged?

How to draw a circuit

Series and Parallel Circuits

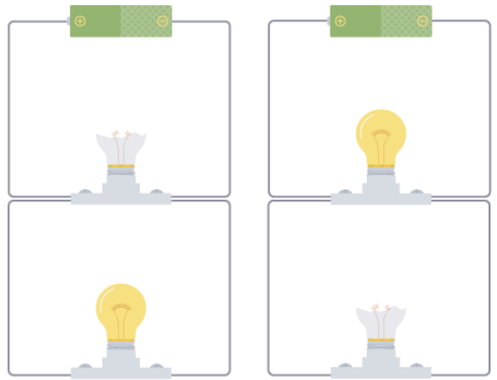
These two bulbs are **in series** with each other.

Current must flow…

These two bulbs are **in parallel** with each other.

Current can only flow through one “loop” of the circuit at a time.

The current is…



What happens if a parallel circuit breaks

Where do we use parallel circuits?

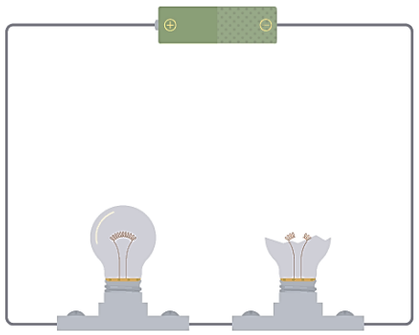
In the parallel circuit shown in this diagram, the two bulbs can be “switched on” separately by closing one of the switches

A diagram showing how a simple circuit is converted into a circuit diagram. It uses straight lines and circuit symbols.

Switch | Cell | Battery

Lamp | Voltmeter | Ammeter

Resistor | Variable resistor | Motor



What happens if a series circuit breaks

Learning Outcomes;	RAG
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